

SmartCell: Rethinking the Project After Review 0

Options and suggestions for addressing faculty feedback — while keeping the core idea (IoT-based 12V lead-acid battery monitoring for petrol two-wheelers) intact.

Faculty Feedback from Review 0

- Battery degradation is already noticeable to riders through symptoms like dim headlights or a weak horn — so the "no visibility" framing needs to be corrected.
- The ML / LSTM prediction layer is unnecessary and adds complexity without real benefit for this problem.
- The project as scoped is too simple and could be completed entirely within Phase 1 — it needs more depth.
- Consider reframing the project around carbon emissions or the UN Sustainable Development Goals (SDGs).

Addressing Each Point Directly

1. "Riders already notice degradation"

This is true, but it is actually a strong argument in your favour rather than a weakness. By the time a rider notices symptoms such as dim headlights or a weak horn, the battery has usually already lost a large share of its usable life — internal resistance typically climbs for weeks before voltage sag becomes noticeable under load. Reframe the pitch from "riders have zero visibility" to:

"Symptoms appear only in the last 10–15% of battery life; SmartCell detects the earlier 85% of degradation that is invisible to the rider."

This is a more defensible claim and keeps the original crux of the project unchanged.

2. "ML / LSTM is unnecessary"

The faculty are right for lead-acid batteries specifically. SOH and RUL estimation for lead-acid is well handled by classical, explainable methods that are already part of the hardware pipeline: internal resistance trend, Coulomb counting, OCV lookup and temperature compensation. Recommendation: remove the LSTM / TensorFlow Lite layer entirely and replace it with a transparent rule-based or statistical degradation model (e.g., curve-fitting internal resistance against a baseline, threshold-based classification). This is more credible for a physical sensing problem and frees up real project time.

3. "Too simple, finishable in Phase 1"

This is the core problem to solve. Removing the unnecessary ML work actually helps here: instead of spending months on a model that was not needed, redirect that effort into a genuine second subsystem — one of the options below adds real additional engineering scope rather than just more slides.

Options for Adding Depth

Each option below keeps the original hardware and target unchanged — a 12V lead-acid battery on a petrol two-wheeler, sensed via ESP32 and reported over BLE to a mobile app. They differ in what additional subsystem or framing is layered on top.

Option A — Active Battery Life Extension (Desulfation Circuit)

Add a desulfation / reconditioning pulse circuit that actively intervenes when early-stage sulfation is detected, instead of only reporting degradation. This turns SmartCell from a passive monitor into an active-intervention device.

- Adds a genuine second subsystem: power electronics and control logic, not just more sensing.
- Fits comfortably within the existing budget (under ₹3,000).
- Directly supports the sustainability narrative — fewer batteries discarded prematurely.
- Gives a clear, demonstrable "before vs after" result for the final review.

Option B — Circular-Economy / SDG Framing (Lead-Acid Waste Reduction)

Lead-acid batteries are a genuine environmental hazard when disposed of improperly — lead and sulfuric acid contamination is a real problem in India's informal recycling sector. Reframe part of the project's outcome around waste and lifecycle impact rather than convenience alone.

- Highlight reduction of premature replacement — batteries scrapped at 60% health when life remained.
- Highlight reduction of unexpected roadside failures, which often lead to rushed, improper disposal.
- Add a simple "battery lifecycle report" in the app: estimated lead-waste or CO2 avoided by extending battery life versus reactive replacement.
- Maps directly to SDG 12 (Responsible Consumption & Production) and SDG 13 (Climate Action) — exactly what faculty asked for, without changing the underlying hardware.

Option C — Charging-System Health → Fuel Efficiency → Emissions Link

A failing battery forces the alternator and voltage regulator to work harder, and a weak charging system or repeated failed cranking attempts burns extra fuel for no result. This option adds a real technical subsystem while producing a quantifiable carbon-emissions story.

- Add alternator charging-efficiency profiling as a new sensed parameter.
- Correlate poor charging efficiency and failed-start events with estimated extra fuel consumption.
- Surface an estimated "emissions impact" metric in the companion app.
- Gives concrete additional engineering work (charging-current analysis) plus a direct climate angle.

Option D — Fleet-Scale Extension

Delivery riders and shared-mobility petrol two-wheelers (food delivery, rental fleets) are exactly where unplanned battery failure causes real economic loss and wasted fuel — a stranded rider often needs recovery or a replacement vehicle. This option extends reach and real-world relevance.

- Add a lightweight fleet dashboard aggregating multiple SmartCell units.
- Flag which vehicles in a fleet need attention, prioritised by predicted urgency.

- Introduces a natural Phase 2 extension and a business/impact dimension.
- Uses the same sensing hardware and app — no pivot required.

Recommended Combination

Combine **Option A** and **Option B**: keep the sensing hardware and companion app exactly as designed, replace the ML block with the simpler classical SOH/RUL algorithm already implicit in the hardware pipeline, add the desulfation circuit as the new hard-engineering component for Phase 1–2, and reframe the Objectives / Impact slides around extending battery lifespan and reducing lead-acid waste as the SDG narrative.

This keeps every existing slide's crux — 12V lead-acid battery, petrol two-wheeler, IoT connectivity, predictive monitoring — intact, while directly answering all three faculty concerns: added technical depth, no unnecessary ML, and a genuine sustainability story.

Quick Comparison

Option	Adds	Effort	SDG / Emissions Fit
A. Desulfation Circuit	Active hardware intervention	Medium	Indirect (extends battery life)
B. Circular-Economy Framing	Reporting / narrative layer	Low	Strong (SDG 12 & 13)
C. Charging → Emissions Link	New sensing + analysis	Medium–High	Strong (direct emissions link)
D. Fleet-Scale Extension	Dashboard, multi-device	Medium	Indirect (efficiency at scale)